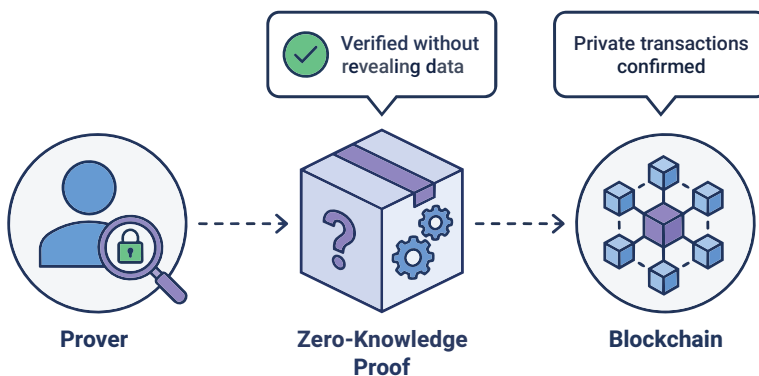


How Zero-Knowledge Proofs Enhance Privacy in Blockchains

Zero-Knowledge Proofs address a fundamental challenge that has existed since the birth of public blockchains: how can a decentralised network verify that the rules are being followed, without making all data public?



is through the classic Ali Baba's Cave example, made famous by cryptographer JeanJacques Quisquater in 1990. Imagine a cave shaped like a ring, with a magic door in the middle that only opens with a secret password. A prover wants to convince a verifier that they know the password, without telling the verifier what the password really is. The prover enters the cave and chooses one of the two paths. The verifier then shouts which path the prover should come out from. If the prover knows the password, they can always come out from the correct path. After many rounds, the verifier becomes convinced the prover knows the password — without ever learning what it is.

Three core properties

Every valid ZKP must satisfy three mathematical properties:

- **Completeness:** If the statement is true, an honest prover can always convince an honest verifier.
- **Soundness:** If the statement is false, no dishonest prover can trick the verifier into accepting it, except with a very small (negligible) probability.
- **Zero knowledge:** The verifier learns absolutely nothing from the proof except that the statement is true. No extra information leaks out.

Interactive vs non-interactive ZKPs

The original ZKPs described by Goldwasser, Micali, and Rackoff in 1989 were interactive. This means the prover and verifier had to communicate back and forth many times. While theoretically solid, interactive proofs are not practical for blockchains, where

When Satoshi Nakamoto launched Bitcoin in 2008, the goal was to create a trustless, decentralised payment system. The solution was a public ledger, the blockchain where every transaction is visible to anyone in the world. This transparency prevents fraud and double spending because no one can quietly alter the record.

However, this same transparency is also a major weakness. In Bitcoin, every wallet address and every transaction amount is public. While wallet addresses are pseudonymous (they do not directly show your real name), researchers have shown that advanced analysis can often link addresses back to real identities. This is not just a Bitcoin problem. Ethereum, the world's most popular smart contract platform, is also fully transparent. Every smart contract interaction, every token transfer, and every wallet balance can be seen by anyone.

For businesses, this is a serious problem. A company using a blockchain to manage supply chain records or financial settlements does not want competitors to see all its transaction

details. For individuals, a transparent blockchain means that financial history, which is normally private, becomes completely public.

Zero-Knowledge Proofs (ZKPs) offer a solution. First described in a landmark 1989 paper by Goldwasser, Micali, and Rackoff, ZKPs allow a prover to convince a verifier that a statement is true without revealing anything else. Applied to blockchain, ZKPs can allow transactions to be verified as valid without revealing the sender, receiver, or amount. Let's explore how ZKPs achieve this, and how they are being used in real open source blockchain projects today.

What are Zero-Knowledge Proofs?

The basic idea

A Zero-Knowledge Proof is a cryptographic method by which one party (called the prover) can prove to another party (called the verifier) that a certain statement is true, without revealing any information beyond the fact that the statement is indeed true.

A simple way to understand this